AADITYA SOGANI
26MAG0064

```
import numpy as np
import matplotlib.pyplot as plt

# ----------------------------
# Parameters
# ----------------------------
N = 200000   # Number of simulations

# ----------------------------
# Example 1: Markov Inequality
# X ~ Exponential(λ = 0.5)
# Mean = 2
# ----------------------------
X = np.random.exponential(scale=2.0, size=N)

a = 5

empirical_markov = np.mean(X >= a)
markov_bound = np.mean(X) / a

print("=== Markov Inequality ===")
print(f"P(X >= {a}) Empirical : {empirical_markov:.6f}")
print(f"Markov Bound E[X]/{a} : {markov_bound:.6f}")
print()

# ----------------------------
# Example 2: Chebyshev Inequality
# X ~ Normal(μ=120, σ=25)
# ----------------------------
X = np.random.normal(loc=120, scale=25, size=N)

mu = np.mean(X)
sigma = np.std(X)

k = 1.8
```

```
empirical_chebyshev = np.mean(np.abs(X - mu) >= k * sigma)
chebyshev_bound = 1 / (k ** 2)

print("=== Chebyshev Inequality ===")
print(f"P(|X-μ| >= {k}σ) Empirical :
{empirical_chebyshev:.6f}")
print(f"Chebyshev Bound : {chebyshev_bound:.6f}")
print()

# ----------------------------
# Example 3: Chernoff Bound
# Sum of Bernoulli variables
# ----------------------------
n = 120
p = 0.65

samples = np.random.binomial(1, p, size=(N, n))
S = np.sum(samples, axis=1)

mu = n * p

delta = 0.25
target = (1 + delta) * mu

empirical_chernoff = np.mean(S >= target)
chernoff_bound = np.exp(-(mu * delta**2) / (2 + delta))

print("=== Chernoff Bound ===")
print(f"Mean μ : {mu}")
print(f"Threshold : {target}")
print(f"Empirical Probability : {empirical_chernoff:.6f}")
print(f"Chernoff Bound : {chernoff_bound:.6f}")

# ----------------------------
# Visualization
# ----------------------------
plt.figure(figsize=(9,5))
```

```
plt.hist(S, bins=40, density=True,
         color='lightgreen',
         edgecolor='black',
         alpha=0.8)

plt.axvline(target,
            color='red',
            linestyle='--',
            linewidth=2,
            label=f'Threshold = {target:.1f}')

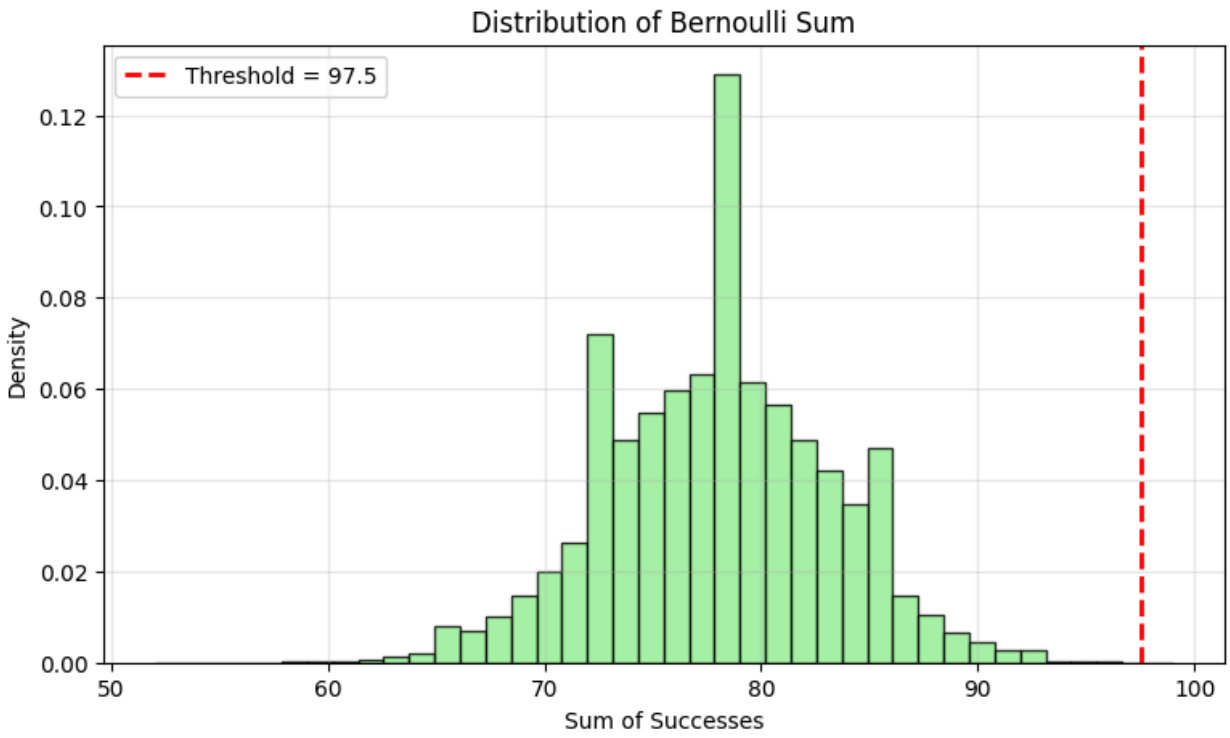
plt.title("Distribution of Bernoulli Sum")
plt.xlabel("Sum of Successes")
plt.ylabel("Density")
plt.grid(alpha=0.3)
plt.legend()

plt.show()
```

```
=== Markov Inequality ===
P(X >= 5) Empirical : 0.081105
Markov Bound E[X]/5 : 0.399443

=== Chebyshev Inequality ===
P(|X-μ| >= 1.8σ) Empirical : 0.071935
Chebyshev Bound : 0.308642

=== Chernoff Bound ===
Mean μ : 78.0
Threshold : 97.5
Empirical Probability : 0.000045
Chernoff Bound : 0.114559
```


Distribution of Bernoulli Sum
Threshold = 97.5
Density
0.00
0.02
0.04
0.06
0.08
0.10
0.12
50
60
70
80
90
100
Sum of Successes